



Lecture 3: System Design

CSC 1202 – Software Development Project



Lecture Objectives



Understand system
architecture



Design a database schema



Understand APIs and
serializers



Connect frontend and
backend

Where
We Are in
the
Course

Week 1: Introduction

Week 2: Requirements

Week 3: System Design

Week 4+: Implementation



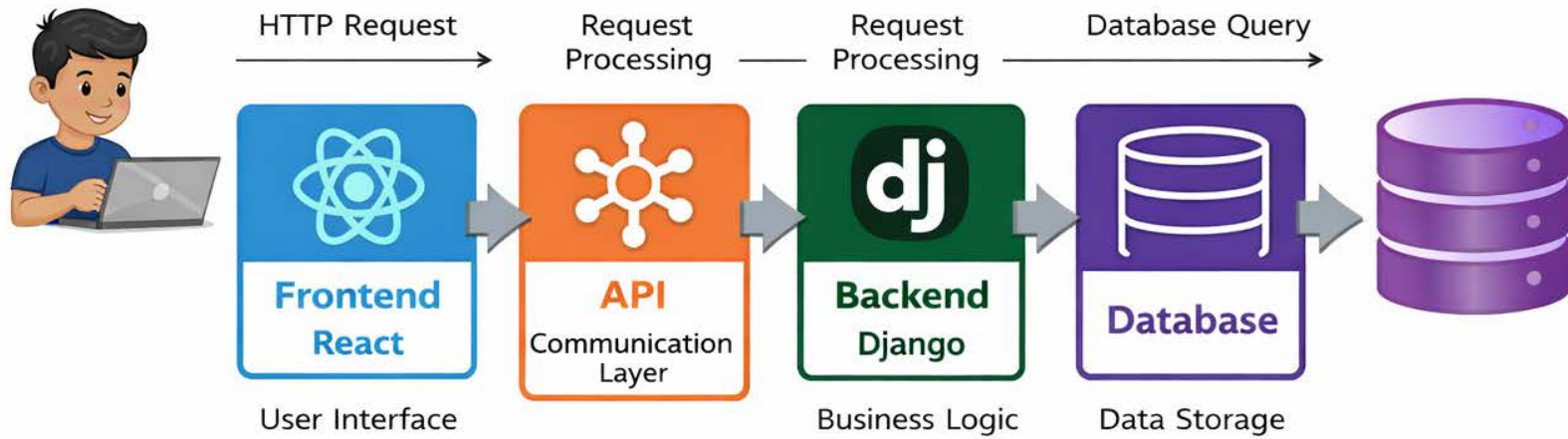
What is System Design?

Defines system structure

Identifies components

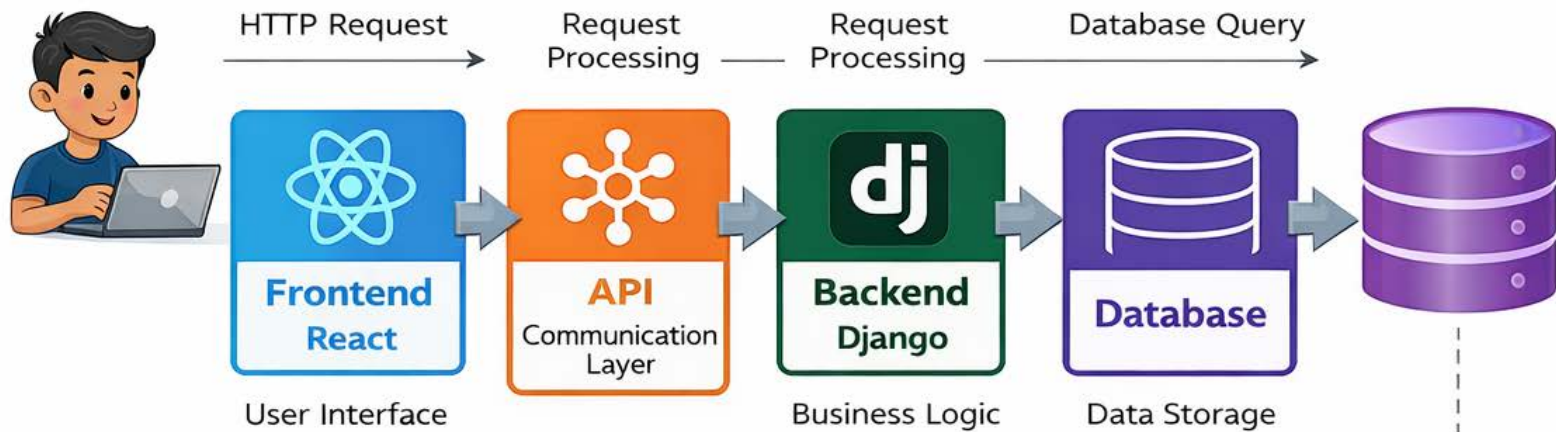
Defines data storage

Defines communication
between parts

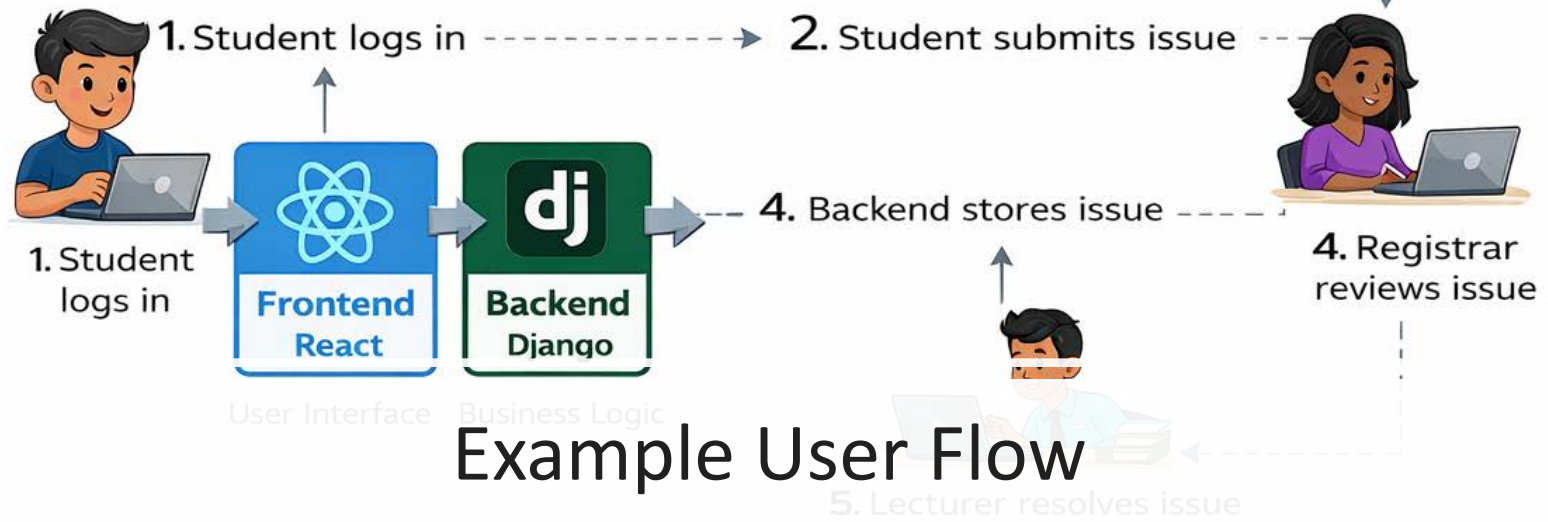


AITs System Architecture

- React Frontend
- Django REST API
- Django Backend
- PostgreSQL Database



Frontend never talks directly to the Database.



Core Entities in AITS

User

Issue

Course

Department

Audit Log

Define Issue

Field	Example
Title	Missing marks for CSC1201
Description	My marks are missing
Course Code	CSC1201
Issue Type	Missing Marks
Status	Open

From User Stories to Design

User story:
Student
submits issue

Identify
nouns →
entities

Define
attributes

Define
relationships

User Story



Entities



Attributes



Relationships



ERD



Code



Issue Entity Example Fields

- title
- issue_type
- course_code
- status
- created_by
- created_at

User Story



Entities



Attributes



Relationships



ERD



Code



Django Model Concept

- A model defines a database table
- Each field becomes a column
- Relationships use ForeignKey

User Story



Entities



Attributes



Relationships



ERD



Code



Model Snippet Example

- `class Issue(models.Model):`
- `title =`
`models.CharField(max_length=200)`
- `issue_type =`
`models.CharField(max_length=20)`
- `created_by = models.ForeignKey(User)`

User Story



Entities



Attributes



Relationships



ERD



Code



API Concept



**Frontend communicates with
backend through APIs**



Example endpoints:

GET /api/issues

POST /api/issues



Serializer Concept

Student submits issue



React sends request



Django API receives request



Serializer validates data



Database stores issue



API returns JSON response

Serializers
convert Django
models to JSON

They also
validate
incoming data



Serializer Snippet

- React cannot understand Django models directly.
- A **serializer** converts **Django objects** into **JSON**.
- JSON is the format used when data is sent through APIs.

</> Python

```
class IssueSerializer(serializers.ModelSerializer):  
    class Meta:  
        model = Issue
```

</> JSON

```
{  
  "title": "Missing marks",  
  "status": "open"  
}
```



View Concept



Views handle HTTP requests



They retrieve data and
return responses



View Snippet

```
GET /api/issues  
POST /api/issues
```

Python

```
queryset = Issue.objects.all()  
serializer_class = IssueSerializer
```



React Fetch Concept

- React sends HTTP requests to APIs
- Fetch data from backend when components load



React Fetch Snippet

```
⌘ JavaScript
```

```
fetch("/api/issues")  
  .then(res => res.json())
```



Displaying Data in React

```
</> JavaScript

issues.map(issue => (
  <p>{issue.title}</p>
))
```

```
Missing marks for CSC1201
Appeal for exam result
Correction of coursework marks
```



Full Request Flow

1. React form submission
2. API request
3. Serializer validation
4. Database storage
5. JSON response

Programming Task This Week

1

Create Issue
model

2

Create
serializer

3

Create API
endpoint

4

Display issues
in React



Key Takeaways

Design before coding

Models represent
database tables

APIs connect frontend
and backend

